

ArithProg

Def: An Airthmetic Progresion is a sequence $a_1, \dots, a_n$ that satisfies the following invariant (It is defined invariantly)

$$\forall n \in \mathbf{Z}, \quad a_{n+2} - a_{n+1} = a_{n+1} - a_n$$

Properties:

From the invariant definition, we have the following properties

Theorem 1: For any arithmetic progression, the common difference is constant. That is, $\forall n \in \mathbf{N} \quad a_{n+1} - a_n = d$ where d is some constant

Theorem 2: Let a_n be an arithmetic progression with common difference d . Then the a_i value of the sequence is given by $a_i = a_0 + i \cdot d$

Theorem 3: Let a_n be an arithmic progression with common difference d , then the sum of values from $\{a_0, \dots, a_k\}$ is given by $\sum_i a_i = k \cdot a_0 + \frac{k(k-1)}{2}d$

Theorem 1: For any arithmetic progression, the common difference is constant. That is, $\forall n \in \mathbf{N} \quad a_{n+1} - a_n = d$ where d is some constant

Proof of Thm 1:

Let a_n be an arithmetic progression. Let $d = a_1 - a_0$ be the common difference between the first two elements of the sequence. Then we must show the common difference is the same for every consecutive element of a_n . That is, we want to show

$$\forall n \in \mathbf{N} \quad a_{n+1} - a_n = d$$

Let $P(n)$ be the claim up to n . We will use induction.

Base Case: $P(0)$ then we must show $a_1 - a_0 = d$

This is true by the definition of d

Inductive Case: Suppose $P(k)$ is true for some partiucular $k \geq 0$. We must show that $P(k+1)$ is also true. That is

$$a_{k+1+1} - a_{k+1} = d$$

Observe, we can use the invariant definition of airthmetic progression to write

$$a_{k+1+1} - a_{k+1} = a_{k+1} - a_k$$

Using the inductive hypothesis on the right hand side, gives the claim. This completes the inductive case $\square$.

Theorem 2: Let a_n be an arithmetic progression with common difference d . Then the a_i value of the sequence is given by $a_i = a_0 + i \cdot d$

Proof: